

TECHNICAL BULLETIN

DIXIE CHEMICAL

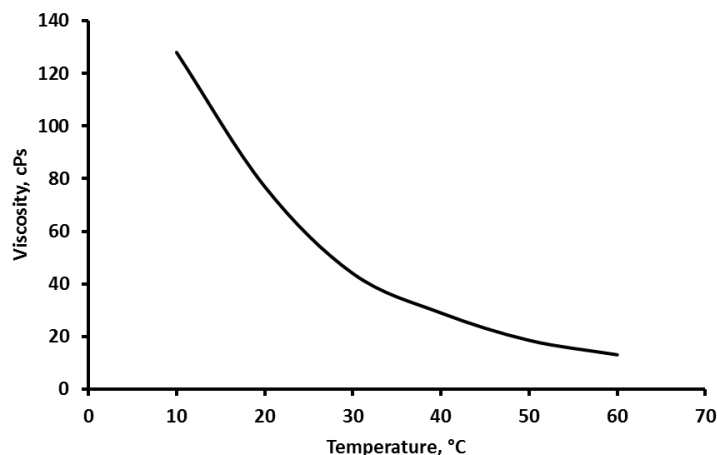
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METHYLHEXAHYDROPHthalic ANHYDRIDE-NC (MHHPA-NC) CAS Reg. No. 25550-51-0

Typical Properties

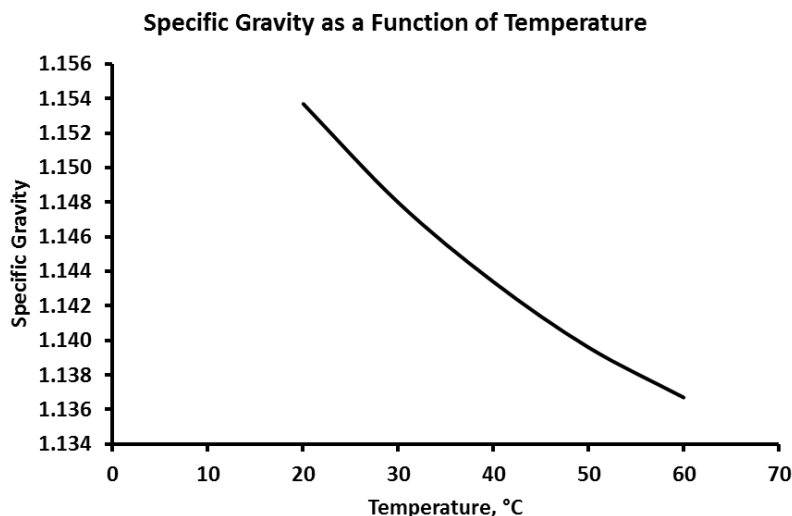
Properties	Value	Test Method
Molecular Weight (average)	168	
Total Anhydride, %	99 Min	D-163
Specific gravity @ 25°C, g/cc	1.15	D-102 (ASTM D4052)
Appearance	Clear Liquid	D-104 (Visual)
Color, Pt-co	25 Max	D-103
Freezing point, °C	-29 to -31	ASTM D1015
Pour point, °F	-38 to -40	ASTM D97

Viscosity as a Function of Temperature



The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

TECHNICAL BULLETIN MHHPA NC



Applications

MHHPA NC is a clear colorless, high purity liquid which can be used alone or in combination with hexahydrophthalic anhydride to produce a lower melting mixture or with other anhydrides to modify physical properties. Its low residual unsaturation level gives maximum resistance to yellowing in coatings and castings, making it desirable where premium optical and electrical performance are needed in epoxy casting and encapsulation applications. Also, gives excellent cure response at cure temperatures, resulting in high crosslinking and relatively high T_g.

For more information on the use of anhydrides like MHHPA NC as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems", available from Dixie Chemical.

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MHHPA NC

Formulation Example

MHHPA NC was formulated with a variety of epoxy resins. The samples were formulated with 100 parts of epoxy resin, 84-117 parts MHHPA NC, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	MHHPA NC, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	85	830	39	156
Low Viscosity BPA Liquid	88	730	37	132
Cycloaliphatic Epoxy	117	112	45	221
Epoxy Phenol Novolac	93	92	32	157
Epoxy BPA Novolac	84	3800	33	142
BPF Liquid Epoxy	95	420	37	148

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

MHHPA-NC will react with water to form methylhexahydrophthalic acid. This is normally undesirable, so MHHPA-NC should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for MHHPA-NC, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

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Special Handling Instructions

In some cases MHHPA NC may crystallize. If crystallization is observed, the material can be heated to return the product to a liquid state.

Loosen bung caps on each drum and place the drums in a hotbox or oven at 80°C for 24 hours. Upon removal from the hotbox, the product should be free of crystals. The product should be used within 48 hours of melting to ensure no reoccurrence of crystal formation.

Heating MHHPA NC following this procedure does not alter the chemical make-up or degrade any properties of the product.

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

MHHPA NC is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.